Pixel 6 support upstream and on android-mainline

William McVicker (Google)
Peter Griffin (Linaro)



東京 2025
LINUX PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

Overview

- Update on Pixel 6 upstreaming progress
  - Where we're at upstream
  - Upstream Testing with Pixel 6
  - Current pain points
- What's next upstream?
- Getting started with Pixel 6 upstream
- Intersection of Android and Pixel 6 upstream
- Questions



Update on Upstreaming Progress

Support added in the past year

- ACPM (pmic, dvfs, clocks)
- Exynos mailbox
- CPU Idle, Hotplug, CPUFreq
- Exynos PMU
- S2MPG1x (RTC, clock) and Maxim 77759 PMICs (nvmem, gpio)
- Many driver clean ups

Functionally, Pixel 6 can boot with a yocto rootfs with the following support:

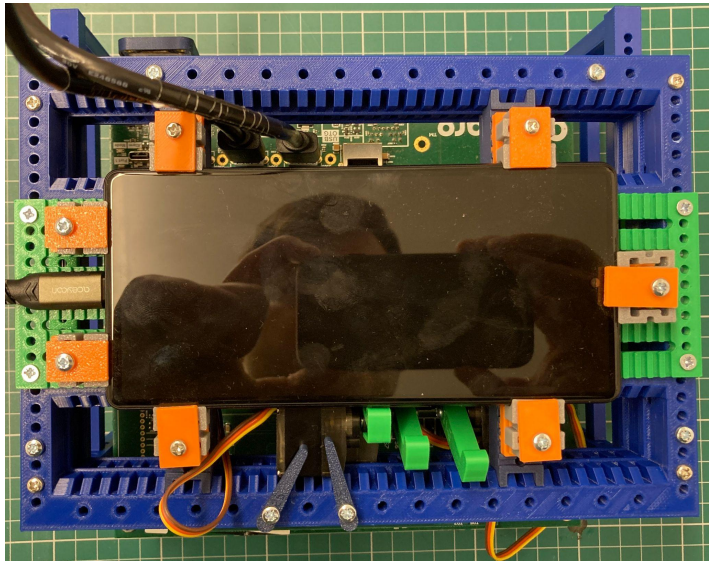
- USB device (enabling adb and fastboot control)
- Android reboot support
  - power off, cold and warm reboot, reboot bootloader, reboot recovery, etc.
- UFS support allowing mounting ext4 partitions including with inline encryption
- simple-framebuffer
- Base PM support (s2idle, cpuidle, cpufreq, ufs hibern8)
- Low speed interfaces (i2c, spi)



Upstream Testing

Upstream Pixel 6 target integrated into LKFT for linux-next testing using the LAVA automation appliance (LAA) https://lkft-staging.validation.linaro.org/scheduler/device_type/gs101-oriole

Boot test recently caught UFS probe Oops in linux-next!



LAVA / Scheduler / Device Types / gs101-oriole — Help

Name	gs101-oriole	Available	1 / 5	Architecture	...
Description	Managed by bakfleet	Retired	2	Processor Family	...
Health Checks	one every 24 hours	Queue	0	CPU model	...
Aliases	...	Running	0	Cores	...

Devices list

Show 25 entries — Search: Press ? for search help

Hostname	Worker Host	Device type	State	Health	Is synced	Last Job	Tags
laa-00038-gs101-oriole	laa-00038	gs101-oriole	Idle	Unknown	✘	—	
laa-00110-gs101-oriole	laa-00110	gs101-oriole	Idle	Unknown	✘	—	
laa-00112-gs101-oriole	laa-00112	gs101-oriole	Idle	Unknown	✘	—	

Health Job Summary

Show 10 entries

Duration	Complete	Failed
24hours	0	0
Week	0	0
Month	0	0

Jobs for devices of type gs101-oriole

Show 25 entries — Search: Press ? for search help

ID	Actions	Device	State	Description	Submitter	Submit time	End time	Duration
199865		laa-00110-gs101-oriole	Complete	TuxTest linaro/lkft#35vN0dQfxXMiSVvyXIU2AZ31P8Q: ltp-smoke@gs101-oriole	tuxtest	Nov.24, 1:06p.m.	Nov.24, 1:32p.m.	0:14:47
199843		laa-00110-gs101-oriole	Complete	TuxTest linaro/lkft#35vN0b8kRiHGnRj9YxssresMfgG: ltp-smoke@gs101-oriole	tuxtest	Nov.24, 1:03p.m.	Nov.24, 1:17p.m.	0:14:26
199829		laa-00110-gs101-oriole	Complete	TuxTest linaro/lkft#35vN0IFdvYJVrotfHLgknCYU2CW: ltp-smoke@gs101-oriole	tuxtest	Nov.24, 12:47p.m.	Nov.24, 1:02p.m.	0:15:08
199654		laa-00110-gs101-oriole	Complete	TuxTest linaro/lkft#35mtZPngSFFJ62FpjdQXDLlaLqS: ltp-smoke@gs101-oriole	tuxtest	Nov.21, 1:11p.m.	Nov.21, 1:50p.m.	0:15:03

Upstreaming Pain Points

Resolved

- Non compliant PSCI firmware requiring special CPU hotplug and CPUIdle implementations
- `of_syscon_register_regmap()` syscon API for custom regmaps
- New `phy_notify_state` API
  - Allows separation of controller glue and PHY ops
- Bootloader DT assumptions, e.g. node names, node hierarchy etc.
- Aligning kernel debug/dev compilation options

Current

- No upstream board overlay support makes it hard to support multiple hardware revisions
- Regulators
  - Multiple PMICs with interdependencies between rails/supplies require changes to regulator core
- Pixel 6 HW wrapped keys implementation depend on Android Keystore HAL
  - Dependency can be removed by refactoring parts of the firmware, but risky to accept for in-market production devices
- Sometimes driver reviews don't start until DT bindings have been reviewed – introducing delay
- Not breaking legacy ARM32 Exynos HW that's unobtainable to test

What's coming next

Plan for the next few releases

- Automatic clocks (on list)
- ChipID (on list)
- TMU (on list)
- TRNG (on list)
- S2MPG1x regulators (on list), ODPM, thermistors, GPIO
- Maxim 77759 fuel gauge & charger (both on list)
- Power Domains (on list)
- IOMMU
- PCIe & PHY
- ACPM full support (ODPM, thermal)
- Display clocks (on list), PHY, panel, DRM/KMS
- Mali (Panfrost)



Introduce support for Pixel 10 (Laguna)

Getting Started with Pixel 6 upstream

Building and Flashing your Pixel 6 with a Yocto rootfs

Follow the the README.md at https://gitlab.com/LinaroLtd/googlelt/pixelscripts to get started

1) Sync the `googlelt/manifest` kernel repo to get the kernel source and build scripts
2) To build and package the kernel images, run the build command:

   ```
   > make images
   ```
3) To use our team's Yocto rootfs, follow the guide and flash the rootfs to the `userdata` partition:

   ```
   > fastboot flash userdata <yocto rootfs>
   ```
4) To flash the kernel and boot with the yocto rootfs, run the command:

   ```
   > make flash
   ```

The device should show up when you run `adb devices` and you should be able to access the device shell prompt with the `adb shell` command.

Intersection of Android and Pixel 6 upstream

Support on android-mainline

- A fully featured Pixel 6 build target has been supported on `android-mainline` for the past few years using out-of-tree drivers forked from the production Pixel 6 kernel branch.
- Allows Android developers to use the latest kernel version to develop and test Android features and contribute upstream.
- Upstreaming Pixel 6 drivers helps
    - reduce tech debt and maintenance of out-of-tree drivers on android-mainline
    - provides a platform to the Android community to upstream their features
    - identify gaps and areas of improvement for future Android development to better support in-market kernel uprevs.
    - bisect regressions more easily
    - earlier testing e.g. linux-next



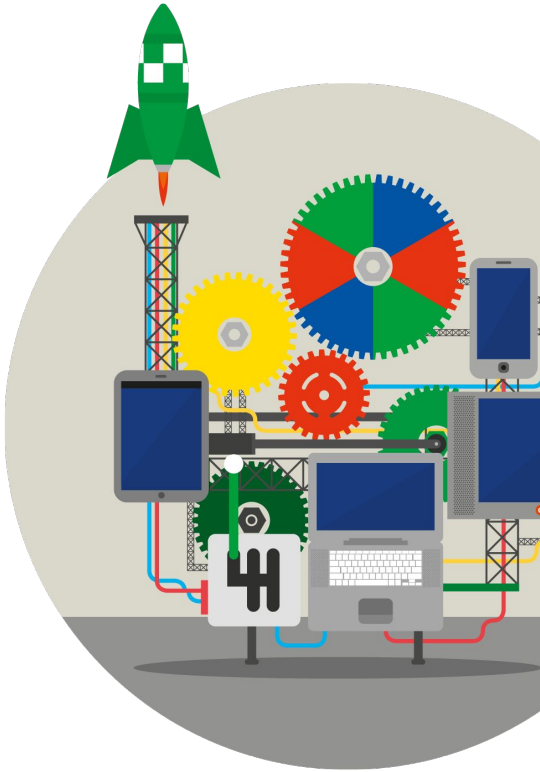
Intersection of Android and Pixel 6 upstream

Pixel 6 upstream target on android-mainline

- The Pixel 6 upstream build target can be compiled from `android-mainline` by following the guide at https://source.android.com/docs/setup/build/building-pixel-kernels
  - Available on https://ci.android.com/builds/branches/aosp_kernel-common-android-mainline/grid
- Supports booting to Android Recovery mode to access the device shell via adb with the upstream kernel drivers.

Challenges integrating the upstream target with Android

- Userspace (Android HAL) incompatibilities
  - Sysfs path differences due to upstream's use of the soc@0 node to organize the MMIO devices together
  - Dependencies on boot-dependent features missing upstream, e.g. hardware-wrapped keys
  - Custom rolled implementations where standardized upstream frameworks exist, e.g. UFS BSG
- SELinux labeling issues due to sysfs path differences
- Integrating gs101 support with older Exynos ARM drivers, e.g. Exynos MCT and TMU drivers, MMIO regmap integration, etc.
- Bootloader firmware assumptions tied to downstream device tree nodes, e.g. UFS, serial, USB naming differences
- Modularization for GKI integration
  - Challenging when drivers are shared between ARM64 and ARM devices due to legacy ARM SoCs lacking the ARM architected timer – need to use Kconfig-fu to allow tristate for ARM64-only configs.
- Upstream's lack of device board overlays requires post-processing for CI testing



Building and Flashing Pixel 6 upstream target from `android-mainline`

Find the detailed guide at https://source.android.com/docs/setup/build/building-pixel-kernels

1) Flash your device to the latest public build with https://flash.android.com
2) Compile the Pixel 6 build target from the android-mainline kernel repo:

```
> bazel run //devices/google/raviole:raviole_upstream_dist
```

3) Flash the kernel images:

```
> fastboot flash boot out/raviole_upstream/dist/boot.img
> fastboot flash --dtb out/raviole_upstream/dist/dtb.img \
    vendor_boot:dlkm out/raviole_upstream/dist/initramfs.img
> fastboot flash dtbo out/raviole_upstream/dist/dtbo.img

# disable AVB verification and wipe your device (only needed once)
> fastboot oem disable-verification && fastboot -w
> fastboot reboot recovery
```

Thanks!
Questions?

